



SUMMARY

Data scientist with strong foundations in machine learning, statistical analysis, EDA, and predictive modeling. Experienced collecting, cleaning, and preprocessing large datasets, developing and evaluating ML models across structured and unstructured data (multi-modal), and deploying production-ready AI systems. Proficient in Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), SQL, and data visualization tools. Published IEEE author with a drive to continuously learn, collaborate effectively across teams, and translate data insights into actionable business solutions.

EDUCATION

University of Pittsburgh, USA

Master's degree in Intelligent Systems (Applied AI)

GPA: 3.73 / 4.00

2023-2025

Nirma University, India

Bachelor's in Instrumentation and control engineering

CGPA: 7.73 / 10

2019-2023

SKILLS

- ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost
- Architectures: CNNs (ResNet50, AlexNet), Transformers, YOLOv8, RF-DETR, CNN-RNN
- Numerical / Data Tooling: NumPy, SciPy, Pandas, Matplotlib, Plotly, Seaborn, ggplot2
- Model Evaluation: Comparative accuracy benchmarking, BLEU scoring, sub-group performance analysis
- Statistical Foundations: Bayesian Networks, Probabilistic Graphical Models, Hypothesis Testing
- Languages: Python, SQL, R, JavaScript
- Dev Tools & Deployment: Git, Jupyter, Google Colab, Streamlit, Flask.

PROJECTS

Courier Management System — Role-Based Database Application

Developed a multi-user database system with role-based access controls (admin, dispatcher, courier, client) using MySQL, Streamlit, Pandas, and Python — practice building and debugging a full software system end-to-end, beyond notebook-level experimentation.

Heart Failure Prediction — Bayesian Modeling & Sub-group Evaluation

Built a Bayesian Network model and evaluated performance across demographic sub-groups using Pandas, Scikit-learn, SciPy, and XGBoost — practice in identifying where and why a model's accuracy degrades on specific slices of data.

RESEARCH & PUBLICATIONS

- Skin Cancer Detection (IEEE Published) : Independently designed and executed a comparative model evaluation on a 10,000-image unstructured dataset (ResNet50, AlexNet, custom CNN); 92% accuracy. Communicated findings through a published research paper. IEEE CISCT 2023
- DOI: 10.1109/CISCT57197.2023.10351329.

PROFESSIONAL EXPERIENCE

Client Services Analyst - University of Pittsburgh | 2023-Present

- Designing an LLM-based candidate-ranking tool that scores student-faculty fit from unstructured inputs like CV summaries returning structured, rationale-backed shortlists to support human-gated, broker-reviewed matching.
- Queried, cleaned, and reconciled large operational datasets in TDX using SQL-style workflows; improved data categorization accuracy by 90% through systematic EDA and data quality analysis.
- Authored process documentation and data workflow reports for cross-functional teams; participated in peer reviews and provided feedback on data entry and categorization standards.

AI consultant (Fellowship)- DMI Aioverse, Pittsburgh | Feb 2025-May 2025

- Collected, cleaned, and preprocessed large-scale satellite imagery datasets via Google Earth API; performed EDA to identify patterns, validate data quality, and engineer features for downstream model training.
- Developed and evaluated object-detection models (YOLOv8 / RF-DETR); conducted structured benchmarking across architectures — RF-DETR achieved 92% accuracy; built annotation and evaluation pipeline reducing manual effort by ~80%.
- Visualized and presented model findings to non-technical stakeholders through scored output dashboards, translating complex ML results into clear, actionable business recommendations.

Machine Learning Intern · DCB India Pvt. Ltd., India Jan – May 2023

- Developed predictive deep learning model (ResNet50 CNN + RNN) for image captioning on unstructured image data; performed EDA, feature engineering, hyperparameter tuning, and BLEU/accuracy evaluation (0.89 / 91%).
- Deployed model to production via Streamlit; produced clear performance reports and visualizations for stakeholder review — demonstrating data quality monitoring and results communication skills.